

1 Problem Statement

Each state $s = (o, h)$ in the state space $\mathbb{S} = \mathbb{O} \times \mathbb{H}$ can be decomposed into two components. Component $o \in \mathbb{O}$ is visible to both the victim and the attacker. Component $h \in \mathbb{H}$ is visible to the attacker, but not to the victim.

The attacker aims to generate a dataset $\mathcal{D}$ by using policy $\pi^\dagger$ to extract data from the environment. The goal is to determine such $\pi^\dagger$ such that training on the dataset $\mathcal{D}$ causes the victim to learn policy π^* that may not be the true optimal policy.

2 Normal Form Games

2.1 Single Observable State

2.1.1 Example

Consider the following reward table.

h	a_1	a_2
0	1	0
1	-1	$-\frac{1}{2}$

Suppose attacker wants victim to learn policy $\pi^* = a_2$. Then attacker can apply $\pi^\dagger(0) = a_2$ and $\pi^\dagger(1) = a_1$. Then the victim would see the following reward table.

a_1	a_2
-1	0

Then the victim will learn $\pi^* = a_2$ based off of the generated data. Note if $\mathbb{P}(h=0) = \mathbb{P}(h=1) = 0.5$, this is not the optimal policy.

2.1.2 Formula

The existence of such a policy $\pi^\dagger = a^\dagger$ when using MLE in this case can be determined using the following.

$$\exists h^\dagger \in \mathbb{H} \left(\forall a \neq a^\dagger \in \mathbb{A} \left(\exists h \in \mathbb{H} \left(r(a^\dagger | h^\dagger) > r(a | h) \right) \right) \right)$$

Here, we apply action $a^\dagger$ in hidden state $h^\dagger$ and obtain reward $r^\dagger = r(a^\dagger | h^\dagger)$. Then for every other action $a \neq a^\dagger$, we find some hidden state h such that applying action a in state h produces a reward that is less than $r^\dagger$.

2.1.3 Considerations

- Increasing the dimension of $\mathbb{H}$ does not matter.
- Maximum likelihood estimate is the root behind this possibility. When generating the dataset, the attacker need only supply one application of

each action. Therefore, the attacker can simply reload the simulation until a desired hidden state is reached.

- The hidden state to apply $a^\dagger$ does not need to be unique. Additionally, if $a^\dagger$ is applied in hidden state $h^\dagger$, other actions $a \neq a^\dagger$ can still be applied in hidden state $h = h^\dagger$.

2.2 Multiple Observable States

Environments with multiple observable states can be reduced to multiple environments, each with a single observable state. Then the calculations for the single observable state can be applied to each subgame.

2.2.1 Correlation

If correlations exist between hidden and observable states, then it is possible for some hidden states to never appear together with certain observable states. Mathematically, we have $\mathbb{P}(h \mid o) = 0$.

To resolve this issue, we add two conditions to the equation derived for the single observable state environments.

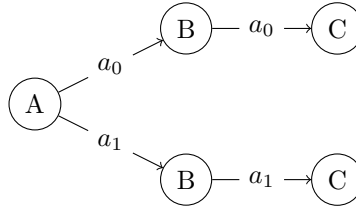
$$\mathbb{P}(h \mid o) > 0 \qquad \mathbb{P}(h^\dagger \mid o) > 0$$

These conditions effectively remove any hidden states that have no chance of occurring when given the observable state.

2.2.2 POMDP Training

After training on the attacker's dataset $\mathcal{D}$, the victim may continue training its policy using POMDP algorithms. The victim could then learn information about the hidden states and deviate from the desired policy π^* .

Imagine the following state transition tree followed by the attacker's policy. Each node corresponds to a state and is labeled with the victim's observation of that state.



In the resulting dataset, these two paths result in the same observations. The dataset will always contain an a_0 followed by another a_0 . It may also contain a_1 followed by another a_1 . However a_0 is never followed by a_1 and a_1 is never

followed by a_0 . Then despite the fact that the observations are the same, the victim may realize this pattern and use it to decipher information about the hidden states.

3 Markov Decision Processes

If the environment is allowed to continue, then the expected future reward must be accounted for. We apply a discount factor γ to allow for infinite horizon games.

3.1 Dataset

Due to the victim's MLE training strategy, the attacker only needs to provide one instance of each observation-action pair.

$$\mathcal{D} = \left\{ \left(o_t, a_t^\dagger, r_t^\dagger, o_{t+1}^\dagger \right)_i \mid i \in \{1, 2, \dots, N\} \right\}$$

3.2 Victim Value Function

For each observable state $o \in \mathbb{O}$, the attacker wants the victim to learn policy $\pi^* = a^\dagger$. The attacker finds some hidden state $h^\dagger$ to apply action $a^\dagger$. Then the victim will expect the reward for state o to be the reward $r^\dagger$ for taking action $a^\dagger$.

$$V^*(o_t) = r^\dagger + V^*(o_{t+1})$$

4 Jerry

I talked to Professor Zhu during his COMP SCI 839 office hours. He said he did not remember discussing this problem, so I tried my best to explain it to him. The example normal form game that I showed him was this.

Observable	Hidden	Action	Reward
a	a	0	2
a	a	1	1
a	b	0	0
a	b	1	-1

The attacker wants the victim to learn action 1 in this state. Then when the hidden state is a , the attacker chooses action 1. When the hidden state is b , the attacker chooses action 0. Action 1 leads to reward 1 and action 0 leads to reward 0. Therefore the victim learns action 1 even though action 0 is clearly the best policy.

He asked about motivation, and I tried to recall the government vs. people example. I don't think I did a very good job though...

I do not remember all of his comments, but here are some of them.

- The motivation behind this problem is not very clear.
- He wasn't too convinced on the connection to reinforcement learning?
- I'm probably not very good at explaining things... (He didn't explicitly state this one though)

I said I will think more and discuss with Professor Wu and may try asking him more if I come up with something.